

Enterprise Network Infrastructure Documentation

Project Overview

This project demonstrates the design and implementation of a secure and highly available enterprise network infrastructure using GNS3. The network connects a Headquarters (HQ) and a Branch Office through a WAN environment, providing reliable communication, centralized services, and secure network access.

Network Technologies Used

Routing & WAN: OSPF, BGP, DMVPN, NAT, Inter-VLAN Routing
Switching: VLANs, EtherChannel
High Availability: HSRP / VRRP
Security: Port Security, BPDU Guard, Root Guard, SSH, ACLs
Server Services: Active Directory and DHCP

Network Structure

HQ VLANs: 10, 20, 30, 40
Branch VLANs: 50, 60, 70, 80

Security Features

Port Security, ACLs, BPDU Guard, Root Guard, SSH Management, and NAT are implemented to secure the infrastructure.

High Availability

HSRP/VRRP and EtherChannel provide redundancy and eliminate single points of failure.

Conclusion

The network is scalable, secure, and resilient, making it suitable for enterprise deployment and advanced networking labs.